

• PORTFOLIO
dylan223.github.io

EDUCATION

B.S. Information Technology, University of Central Florida

Expected Fall 2026

SKILLS

Languages	Lua, GLua, Python, C, C++, Java, SQL, HTML, CSS, JSON
ML & Data	scikit-learn, pandas, NumPy, K-means clustering, PCA, linear regression, LLM integration (Anthropic Claude API)
Tools	MySQL, PhpMyAdmin, Apache, AWS, Django, Git, FileZilla, PuTTY, Source Engine, Hammer, Unity, Unreal Engine 4/5
Systems	Linux (VPS deployment & administration), Windows, VirtualBox, REST APIs
Certifications	CompTIA Network+

PROJECTS

Player Analytics Platform — Machine Learning & LLM Integration

gentoi.online/analytics

- Built an end-to-end **machine-learning analytics platform in Python** over **43,980 players**, ingesting three MySQL schemas across two servers and unifying mismatched Steam ID formats (legacy STEAM_0 ↔ 64-bit) through entity resolution.
- Trained a **K-means clustering** model (scikit-learn) to segment players into five behavioural archetypes; selected cluster count via silhouette scoring and visualized separation with **PCA** projection to 2D.
- Built a **linear regression** predicting in-game wealth from engagement features (R^2 0.448) with standardized coefficients, log-transformed inputs, and a held-out test split; flagged multicollinearity to keep findings correlational.
- Integrated the **Anthropic Claude API (LLM)** with a precompute-and-narrate design so generated insight cites only verified, Python-computed figures; deployed live with per-IP rate limiting and results cached to the production database.

Gentoi.online — Custom Web Platform & Game Server Gateway

gentoi.online

- Designed and built a full-stack web platform from scratch as the operational frontend for an active multiplayer server (4,000+ monthly / 80,000+ lifetime players), with Steam OpenID auth, a transactional store (2,000+ purchases), a public bans registry, and live gameplay stats.
- Architected and tuned the backing database powering every signup, transaction, and stat aggregation; deployed and maintained the full stack independently (frontend, backend, database, and Linux server infrastructure).

Game Player Database — Production SQL Backend

Independent Project (with content creator)

- Built and managed a full-scale MySQL backend scaling to 50,000+ players and millions of rows, with normalized schemas, stored procedures, and import/export pipelines for reliable high-volume data under production traffic.
- Wrote custom Lua scripts to read and update player records in real time for concurrent users; deployed a Linux VPS from scratch (MySQL, Apache, Git, role-based permissions, PhpMyAdmin).

Game Server Administration & Workshop Content

Freelance / Independent

- Sustained 99%+ uptime across multiplayer servers serving thousands of players; built and managed Windows Server VM networks in VirtualBox for multi-server testing and network administration.
- Authored original Garry's Mod NPCs and Nextbots (Source Engine) with custom AI, animation, and effects in Lua/GLua, totaling 300,000+ Steam Workshop subscriptions.

Simple Train — ML Training Backend & Custom Hardware

Hands-On Hardware & Backend

- Developed a Django REST API backend bridging the frontend and an ML training pipeline; built and optimized a dedicated server PC for ML training workloads and resolved thermal, boot, and compatibility issues through structured troubleshooting.

WORK HISTORY

Game / Program Freelancer	Everyday AS	2017 – Present
Operating Systems Support	Onesupport	Jun 2023 – Aug 2023
Host	Red Lobster	Jan 2018 – Jul 2019